

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T03 · Prime Factors, HCF and LCM

Questions

Student book · questions with writing space

■ 14 pages

FAMILY MAP

Prime Factors HCF LCM

F01 Choose the common-prime envelope

9 Q | 19 marks

F02 Build or manipulate prime-power form

4 Q | 11 marks

Question 02

4MA1-1H Q3

MODULAR UNIT 2

2 marks

- 3 Find the highest common factor (HCF) of 72 and 108
Show your working clearly.

Question 03

4MA1-1H Q7 demand

MODULAR UNIT 2

2 marks

7 $A = 5^3 \times 7^3 \times 11^6$ and $B = 5^6 \times 7^2 \times 11^4$

Find the highest common factor (HCF) of A and B

Give your answer as a product of powers of its prime factors.

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Question 07

4MA1-1H Q8

MODULAR UNIT 2

2 marks

- 8 (a) Write 300 as a product of its prime factors.
Show your working clearly.

(2)

Question 08

4MA1-1H Q8

MODULAR UNIT 2

2 marks

$$A = 2 \times 2 \times 2 \times 3 \times 3 \times 5$$

$$B = 2 \times 2 \times 3 \times 3 \times 3 \times 5$$

- (b) Find the lowest common multiple (LCM) of $5A$ and $7B$
Show your working clearly.

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(2)

Question 05

4MA1-1H Q6 M01

MODULAR UNIT 2

2 marks

6 $A = 2^3 \times 5^4 \times 7 \times 11$

$B = 2^2 \times 5^2 \times 7^2$

$C = 2^2 \times 5^3 \times 7^4$

Find the highest common factor (HCF) of A , B and C
Write your answer as a product of prime factors.

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Question 03

4MA1-1H Q3 Q3(a)

MODULAR UNIT 2

2 marks

- 3 (a) Find the highest common factor (HCF) of 56 and 84
Show your working clearly.

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(2)

Question 04

4MA1-1H Q3 Q3(b)

MODULAR UNIT 2

2 marks

- (b) Find the lowest common multiple (LCM) of 60 and 72
Show your working clearly.

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(2)

4MA1-2H Q7

3 marks

$$\begin{aligned} A &= 2^3 \times 3^2 \times 5^2 \times 11 \\ B &= 2^4 \times 3 \times 5^4 \times 13 \end{aligned}$$

Find the lowest common multiple (LCM) of A and B .
Give your answer as a product of powers of prime numbers.

$$B = 16 \times 3^{37}$$

(1)

(b) Express the number $A \times B$ as a product of powers of its prime factors. Give your answer in its simplest form.